SW·AI LEADERS PROGRAM · TEAM B

Prototype Plan

Traffic Signal Optimization with Mobility Data and AI — a SUMO-based signal comparison simulator

October 28, 2026 · Optional — not required by the program · First version (MVP)

In one line

A simulator that compares fixed-time signals, Google-style "AI recommends → engineer approves," and Busan-style "real-time automatic control" on the same virtual road, and shows how the results change with navigation-data coverage and in emergencies or disasters.

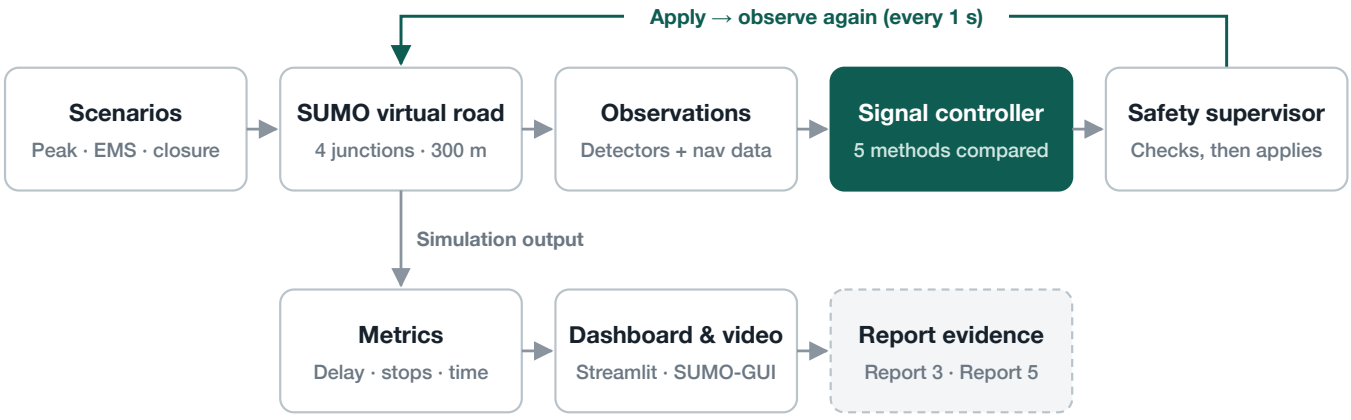
Why build a prototype (optional)

- The program doesn't require a prototype. The official output is five activity reports (Research & Plan → Interviews → Problem definition → Planning with CSUEB → Review & results sharing), which is closer to a proposal.
- We don't have to build one, but we think it would be worth it: results we've actually run would make our "one problem + one solution" in Report 3 much stronger.
- The base is the four-intersection SUMO testbed from Aditya's research report and role packages. The original design is research-project scale, so the first version is scoped down.
- AI tools can be used freely for research, organizing and coding.

How the prototype tests our questions

Core question from Report 1	Prototype experiment
1 Sampling bias Is a signal plan built only from navigation-app vehicles good enough?	Vary navigation-vehicle coverage (10, 25, 50, 100%) and compare how much the AI recommendation and the metrics change
2 Spillover Does fixing one intersection clog the next one?	Connect four intersections and, in the road-closure scenario, measure how long and how far queues spread to the neighboring intersections
3 Measuring impact Can we measure the effect without weather or weekday effects?	Change only the signal method under the same traffic conditions (same seeds) → a clean A/B comparison with no outside factors

System overview



Five signal methods we'll compare

1 **Fixed-time** The same timing every 90 s cycle. Our starting point. — Baseline

2 **Actuated** Extends green when detectors see waiting cars (built into SUMO). — Baseline

3 **Max-pressure** Gives green to the direction with the biggest gap between incoming and outgoing queues. No learning. — Strong baseline

4 **AI recommender (Google-style)** Searches signal timing automatically in the simulation (Optuna), then applies it after an engineer approves. — Core

5 **Reinforcement learning (Busan-style)** AI decides the signals directly in real time (SUMO-RL). — Stretch

Principle Following Aditya's report, we won't compare AI only against a deliberately weak fixed-time baseline. Strong baselines like methods 2 and 3 are included.

Tech stack

Area	Tool	Use
Simulation	Eclipse SUMO netconvert · SUMO-GUI	Build the road network and signal programs, record the screen
Control & execution	Python + TraCI	Step the simulation every second and control the signals
AI recommender	Optuna	Automatically search cycle length, green splits and offsets between intersections in the simulation
RL (stretch)	SUMO-RL	Real-time AI signal control
Data & metrics	pandas	Navigation-vehicle sampling, metric calculation and comparison
Demo	Streamlit	Pick a scenario and a signal method to compare the metrics
Collaboration	GitHub · AI coding tools	Share code and move faster (no limits on AI use)

Metrics

Mean delay (primary) · Stops · Travel time (mean, p95) · Queues & spillback · Unfinished trips · Emergency vehicle arrival time

We always report unfinished trips too, so a controller can't look better just by keeping cars from finishing their trips (a principle from Aditya's report).

Test scenarios

Baseline · Rush hour

Heavy traffic on the east–west arterial.

Measure Mean delay, stops, travel time

Emergency vehicle priority

Aditya's idea

When an ambulance or fire truck approaches, give green in its direction.

Measure Emergency vehicle arrival time, extra delay for other vehicles

Road closure

Aditya's idea · flood/sinkhole

One road is blocked and detour traffic piles up.

Measure How long and how far congestion spreads to nearby intersections, recovery time

Security test

Yunseong

Check whether fake navigation data fools the AI recommender and whether the safety supervisor filters it out. Inspired by the 2020 Berlin case where an artist used 99 phones in a handcart to create a fake traffic jam on Google Maps.

Aditya's other scenarios (earthquakes, landslides and more) will be added once the first version works.

Roles

Joshua (Seong Hyeon) Kim

Software & integration

Road network and signal programs, TraCI runner, safety supervisor, dashboard integration

Existing role: team lead, global cases, Google questions, final presentation

Doyoung Lee

Data & evaluation

Navigation-vehicle sampling, metric calculation, result analysis

Existing role: effectiveness metrics and data analysis, Databricks questions

Yunseong Lee

AI & security

Max-pressure, reinforcement learning, security tests

Existing role: Korean cases (Jeju, Busan), data security and privacy, Tesla questions

Aditya Tirunagaru

Scenarios & analysis

Proposed

Emergency and disaster scenario design, impact and cost-benefit analysis (with Doyoung), and optionally an API that shares signal data with navigation apps

Official part: the CSUEB section of Report 4 (Planning & brainstorming)

Timeline

By Oct 31 · Before departure

Build the baseline

Road network + fixed-time and actuated signals + metric extraction → baseline numbers

Nov 2–4 · Site visits

Add the AI recommender

Reflect what we learn at Google, Tesla and Databricks

Nov 5 · CSUEB

Connect the scenarios

Add emergency and disaster scenarios with Aditya and use them in the Report 4 brainstorming

Report 5 · Results sharing

Demo

Share the results with the dashboard and recordings

> **Later** · MPC (predictive control), multi-agent RL across intersections, automated tests (CI) and 20-run statistical analysis come after the first version works. Aditya's documents also describe this design as proposed assumptions that haven't been validated yet.

How it connects to the activity reports

Activity report	What the prototype adds
Report 2 · Interviews	Questions that come up while building, asked during the visits (e.g., how does Google correct for low navigation-data coverage?)
Report 3 · Problem definition	"AI adoption review" = AI recommender results · "Human input review" = engineer approval and the safety supervisor
Report 4 · Planning	CSUEB section = results from Aditya's emergency and disaster scenarios
Report 5 · Review & sharing	Demo with the dashboard and recordings

References

- Aditya Tirunagaru — Team B Comprehensive Traffic AI Research Report (Oct 2026) and three role packages
- Team B Activity Report 1: Research & Plan (revised Oct 15, 2026)
- Google Research — Project Green Light
- Eclipse SUMO documentation
- SUMO TraCI (Python control interface)
- SUMO-RL (RL environments for signal control)
- Optuna (automatic optimization framework)
- Streamlit (data dashboards)
- Tom's Guide (Feb 2020) — 99 smartphones create virtual traffic jam in Google Maps